Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	08 March 2026
Team ID	—
Project Name	Predictive Pulse: Harnessing Machine Learning for Blood Pressure Analysis
Maximum Marks	4 Marks

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web-based form where users enter health parameters such as age group, symptoms, blood pressure levels, and medication history	HTML5, CSS3
2	Web Server	Handles HTTP requests from the frontend and processes form submissions	Python Flask (app.py)
3	Backend Processing	Validates input data, converts categorical features into numerical format, and prepares data for prediction	Python, NumPy
4	Machine Learning Model	Predicts hypertension stage based on input health parameters	Scikit-learn Logistic Regression
5	Model Storage	Stores trained machine learning model for real-time prediction	Joblib (.pkl model file)
6	Recommendation Engine	Generates medical recommendations and lifestyle advice based on predicted risk level	Python Logic in Flask
7	Data Input Features	Health-related attributes such as age group, gender, symptoms, blood pressure readings, and medication history	Structured Form Data
8	Output Display	Shows prediction results including hypertension stage, risk level color indicator, confidence score, and recommendations	HTML + Flask Template Rendering
9	Development IDE	Used for coding, debugging, and managing project files	Visual Studio Code
10	Infrastructure	Local development environment for running and testing the application	Python Virtual Environment (venv)

Table 2: Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Framework	Entire project uses open-source technologies without licensing cost	Python, Flask, Scikit-learn
2	Security	Input validation implemented to prevent invalid data submission	Flask Input Validation
3	Scalable Architecture	Modular design allows easy replacement of machine learning models or addition of new health parameters	Flask + ML Model Integration
4	Availability	Application runs locally through a web browser and can be deployed on cloud platforms	Local Server, Future Cloud Deployment
5	Performance	Lightweight machine learning model ensures quick predictions and fast response time	Logistic Regression + Flask